ThoughtFlow-FP8: A Recurrence-Distance Gate for Sparse KV Retention

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Abstract

ThoughtFlow-FP8 tests whether training-free sparse KV retention can preserve useful reasoning context better than matched-budget KV-retention proxies. Synthetic traces support the intuition that phase markers and anchors can be protected, but real retained-context quality and a larger sparse-cache gate rule out further tuning of the original anchor/recent/phase/math family on the available traces. We therefore pre-registered one successor signal, recurrence-distance utility (`rdu_topk`), which ranks prefix tokens by delayed self-attention reuse across lag buckets and uses no token labels, recent reserve, or continuation loss. On the frozen 74-trace distilgpt2 sparse-cache gate at a 0.20 keep fraction, `rdu_topk` reaches NLL 3.779 versus ThinKV-like 3.900 and R-KV-like 3.939, with paired deltas -0.121 [-0.211,-0.037] and -0.160 [-0.264,-0.050]. A measured no-retuning same-slice rerun reproduces the cached gate exactly on the current deterministic Mac stack, while cached deterministic half-splits keep positive mean margins in 4/4 partitions but only 2/4 partitions clear both paired confidence intervals below zero. This revives the branch on the Mac-local decision surface, but it is not yet an ICLR-ready positive method without a larger or independently seeded reproduction.

1 Policy Family

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. Those policies are now stopped on the current saved traces. The only live successor in this shell is `rdu_topk`, a pre-registered recurrence-distance utility policy based solely on prefix self-attention.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 1: Retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

2 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor	Phase	Math-state
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
`value_norm_topk`	0.207	0.000	0.492	0.845
train-selected sweep	0.207	1.000	0.430	0.956

Table 2: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is `value_norm_topk`. ThoughtFlow beats it on phase recall, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against `value_norm_topk`, ThoughtFlow's phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a reviewer-facing claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

3 CPU Sparse-Cache Quality Gate

Finally, we ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result, but it is stronger than retained-text NLL because the model actually receives a sparse cache.

Policy	Keep rate	NLL	Paired Δ vs R-KV-like
full cache	1.000	2.142	-1.296 [-1.533,-1.066]
ThoughtFlow-saliency-recent	0.210	3.372	-0.067 [-0.151,+0.011]
ThinKV-like	0.210	3.389	-0.049 [-0.192,+0.077]
ThoughtFlow-recent	0.210	3.399	-0.040 [-0.169,+0.074]
ThoughtFlow sweep best	0.210	3.436	-0.002 [-0.007,+0.000]
R-KV-like	0.210	3.438	0.000
LongFlow-like	0.210	3.588	+0.150 [-0.016,+0.300]
ThoughtFlow	0.210	3.588	+0.150 [-0.006,+0.297]

Table 3: CPU sparse-KV continuation quality on 24 distilgpt2 traces. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the pre-registered 0.03 promotion margin.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects `tf_sparse_r0.55_p0.05_m0.12_a2` on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a positive result.

We then froze `ThoughtFlow-saliency-recent` and `tf_sparse_r0.55_p0.05_m0.12_a2` and ran a larger 74-trace sparse cache slice with no retuning. ThinKV-like is the best compressed row among the stopped family comparison at NLL 3.900. The frozen sparse ThoughtFlow row gets 3.908, with paired delta -0.031 versus R-KV-like and 95% CI [-0.078,+0.020], but +0.008 versus ThinKV-like with 95% CI [-0.060,+0.085]. ThoughtFlow-saliency-recent gets 3.920. The larger no-retuning gate therefore stops this policy family.

After that stop decision, we pre-registered exactly one successor, `rdu_topk`. It computes recurrence-distance utility from full-prefix prefill attentions using lag buckets [8,15], [16,31], [32,63], and [64,∞], then keeps the top budget by utility with ties broken by lower position. It does not use continuation tokens, trace labels, recent reserves, anchors, phase markers, or math-state bonuses.

Policy	Traces	Keep rate	NLL	Paired Δ vs ThinKV-like
full cache	74	1.000	2.848	-1.052 [-1.199,-0.919]
`rdu_topk`	74	0.213	3.779	-0.121 [-0.211,-0.037]
ThinKV-like	74	0.213	3.900	0.000
frozen sparse ThoughtFlow	74	0.213	3.908	+0.008 [-0.060,+0.085]
ThoughtFlow-saliency-recent	74	0.213	3.920	+0.020 [-0.030,+0.074]
R-KV-like	74	0.214	3.939	+0.039 [-0.015,+0.099]
LongFlow-like	74	0.213	4.158	+0.258 [+0.178,+0.337]

Table 4: One-shot frozen sparse-cache successor evaluation. `rdu_topk` also beats R-KV-like by -0.160 NLL with 95% CI [-0.264,-0.050], clearing the pre-registered promotion rule.

Cached split diagnostic. We then reused the same 74-trace rows without rerunning the model or changing the `rdu_topk` rule. Even/odd and first-half/second-half partitions all keep `rdu_topk` as the best compressed row and preserve at least 0.03 mean NLL margin versus both R-KV-like and ThinKV-like. However, only 2/4 half-size partitions keep both paired confidence intervals below zero. This supports the current promotion decision but does not replace an independent larger or seed-repeated reproduction.

Measured no-retuning rerun. We also reran the frozen probe on the same 74 traces and wrote a separate measured artifact rather than overwriting the cached gate. On the current deterministic Mac stack, all policy NLLs match the cached promoted gate exactly: `rdu_topk` remains best compressed at NLL 3.779, with zero measured minus cached NLL drift. Strict separation is preserved: measured margins versus `rdu_topk` are +0.141 and +0.129 NLL for the stopped ThoughtFlow rows, and +0.160, +0.121, and +0.379 for R-KV-like, ThinKV-like, and LongFlow-like. The measured per-trace compressed oracle reaches NLL 3.634, leaving `rdu_topk` 0.145 NLL above that oracle and 0.931 NLL above full cache, with a 0.419 oracle-hit rate. This is useful reproduction bookkeeping, not a new independent slice.

4 Decision

The branch is revived only for the pre-registered recurrence-distance successor. The stopped anchor/recent/phase/math policy family remains ruled out on this trace set. A paper-ready claim now requires evaluation-quality work: larger frozen slices, seed repeats, and then competitor and long-context benchmarks if the `rdu_topk` signal survives the same strict same-family, cross-family, and oracle/headroom reporting.

5 Artifacts

- `experimental/thoughtflow_fp8/phase2/perplexity_impact_proxy.md`
- `experimental/thoughtflow_fp8/phase2/policy_sweep.md`
- `experimental/thoughtflow_fp8/phase2/hidden_saliency_retention_probe.md`
- `experimental/thoughtflow_fp8/phase2/kv_drop_quality_probe.md`
- `experimental/thoughtflow_fp8/phase2/frozen_sparse_cache_probe.md`
- `experimental/thoughtflow_fp8/phase2/rdu_robustness_diagnostic.md`

- `experimental/thoughtflow_fp8/phase2/rdu_no_retune_reproduction_check.md`
- `experimental/thoughtflow_fp8/phase2/real_trace_retention_sweep.md`
- `experimental/thoughtflow_fp8/paper/colm_workshop_shell.md`